Fiscal Year:	FY 2004	**Task Last Updated:**	FY 04/03/2006
PI Name:	Sacksteder, Kurt R. Ph.D.		
Project Title:	Flame Spread and Extinction in Partial Gravity Environments		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Surface combustion, shs, fire safety and smoldering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Cleveland	**State:**	OH
Zip Code:	44135	**Congressional District:**	10
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	10/01/1998	**End Date:**	09/30/2003
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	T'ien, James (Case Western Reserve University) Ferkul, Paul (National Center for Microgravity Research)		
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This task consists of both experimental and theoretical efforts. Flame spreading experiments will be conducted, burning thin fuels in purely-buoyant flows in partial-gravity aboard reduced-gravity aircraft facilities, mainly using existing equipment. An extensive numerical modelling effort will also be conducted to provide simulations of flame spread in purely buoyant and purely forced flows. The purpose of this research effort is to examine flame spreading and material flammability in partial-gravity environments, extended as part of this effort to as low as 10-3 g/g Earth, with the objectives: 1) to determine the flame structure, flame spreading and material flammability characteristics of thin fuels burning in opposed, buoyant flows (downward spreading); 2) to determine the flame structure, flame spreading and flammability characteristics of thin fuels burning in concurrent, buoyant flows (upward spreading) over the entire expanded range of partial-gravity levels;		

	3) to identify the similarities and differences of flame spreading over simple fuels in forced versus buoyant flows to aid in predicting flammability limits and flame spread behavior in extraterrestrial environments; and 4) to advance the technology for conducting high-quality reduced-gravity aircraft experiments. This effort will provide the first experimental data and theoretical background on the relative flammability of materials and corresponding fire hazards that will be encountered in Lunar and Martian environments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)